DisaggKV: Scalable and Disaggregated CXL-PIM Pooling for Multi-Tenant LLM Serving

Rearchitecting Multi-Tenant LLM Serving Interconnects

Kaichen Li

March 31, 2026

Department of Computer and Information Engineering
Youngsan University, South Korea

Introduction: The LLM Memory Challenge

- **Explosive Demand**: Multi-tenant LLM serving pushes infrastructures to limits.
- **KV Cache Footprint**: 7B model @ 128K context consumes >**7GB** per request.
- **HBM Exhaustion**: Local GPU HBM is rapidly exhausted by concurrent batches.
- **CXL Solution**: Enables physical memory pooling, but introduces performance degradation during autoregressive decoding.

Motivation: The Throughput Bottleneck

- **Host-Aggregated CXL**: Standard CXL 2.0 pools rely on Host to fetch all data.
- **Data Movement**: Fetching dense KV vectors across Host-indirected CXL fabric saturates PCIe bandwidth.
- **SLA Violation**: Violates strict 99% tail-latency (e.g., 50ms) Service-Level Agreements.

The Problem
Repeatedly fetching dense historical KV vectors to resolve Attention dot-products creates a severe I/O congestion wall.

The Queue Congestion Wall: M/D/1 Theory

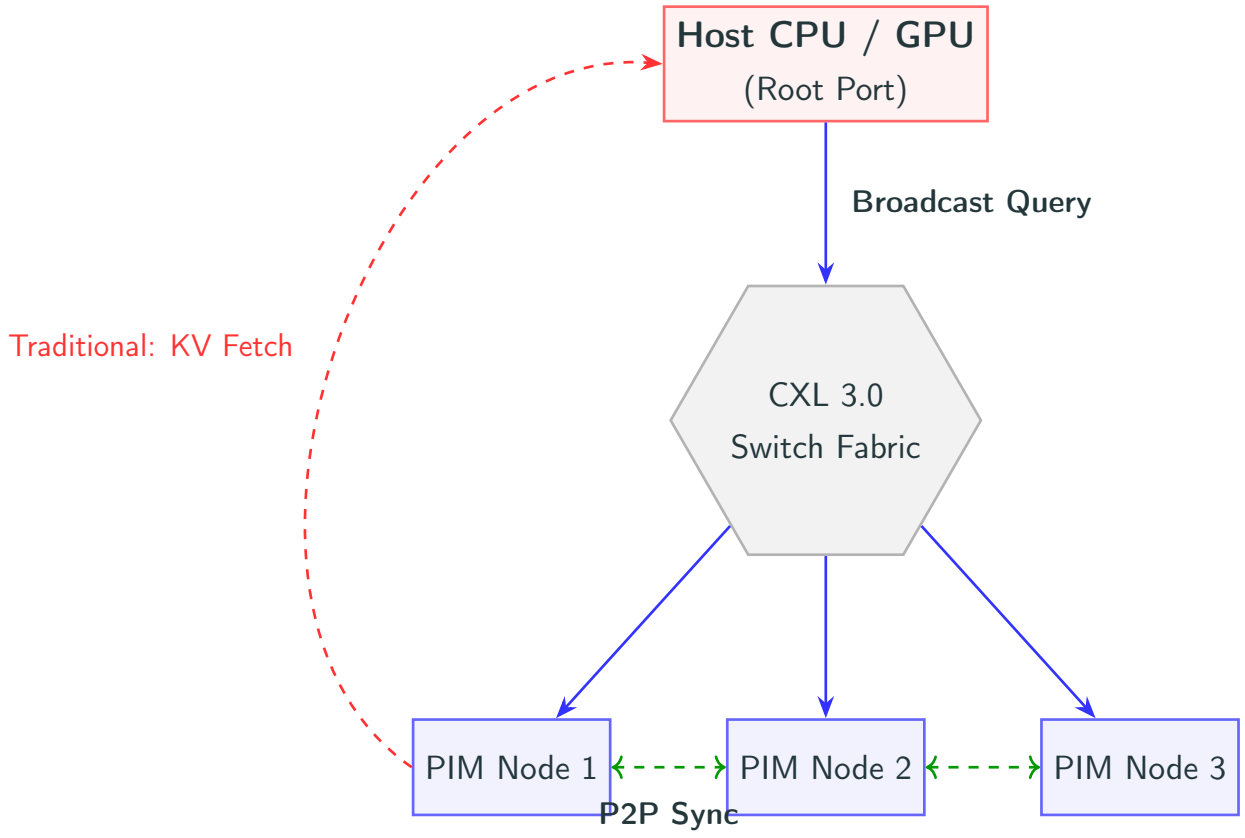
- Expected delay W_q in a deterministic network port:

$$W_q = \frac{\rho}{2\mu(1-\rho)} \quad \text{where} \quad \rho = \frac{\lambda}{\mu}$$

- λ **(Arrival rate)**: Increases with distributed KV subsets.
- μ **(Service rate)**: Restricted by PCIe Gen6 x16 (64GB/s).
- As $\rho \to 1$, queue latency grows asymptotically.

Strategy: Suppress arrival density λ at the central node by keeping computation co-located at endpoints.

DisaggKV: System Overview



Transforming interconnect from congested data-fetch bus into reduction fabric.

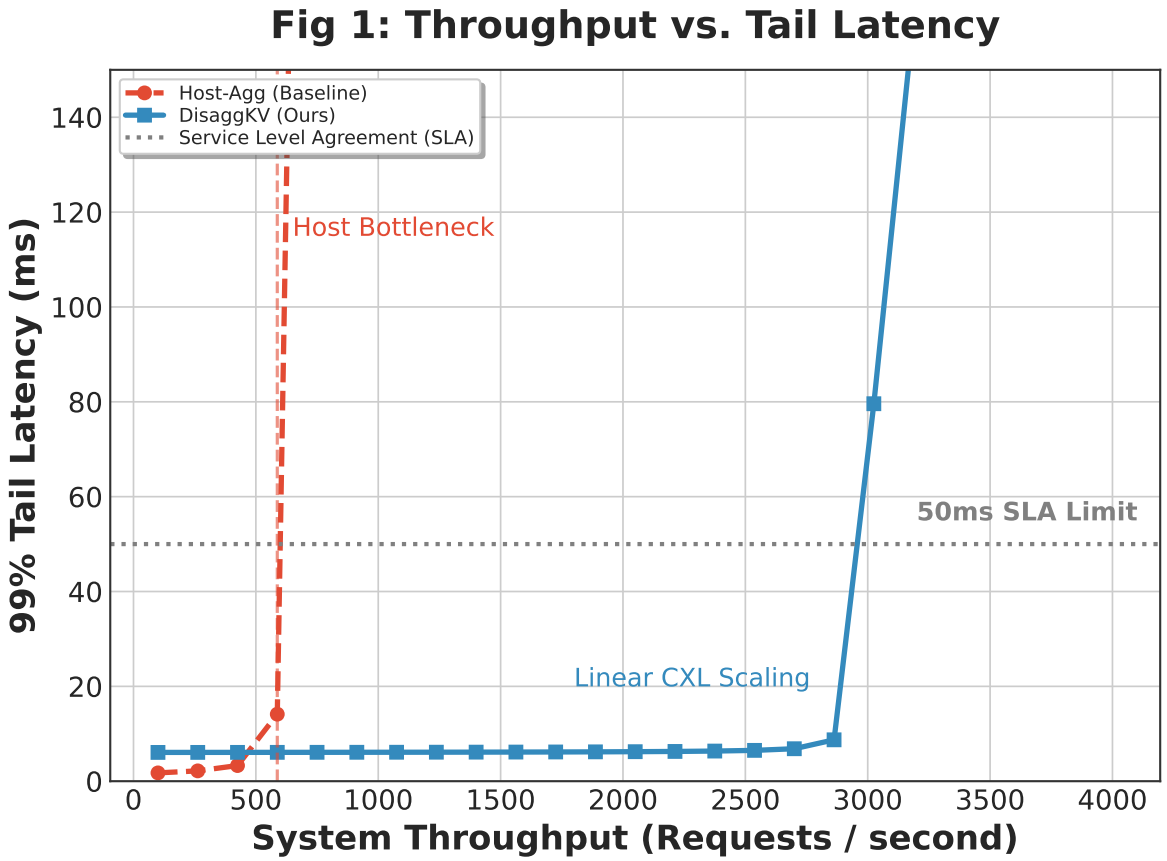
Core Contribution 1: Locality-Aware Paging

- **Segregation**: Shared Hot Pages (e.g., System Prompts) vs. Private Contexts.
- **Pinned Prompts**: High-contention prompts pinned to dedicated CXL nodes.
- **Dynamic Migration**: Monitor CXL switch queue depths; migrate blocks to idle nodes to ensure balanced load (ρ).
- **Benefit**: Prevents redundant read amplification and balances bandwidth.

Core Contribution 2: Fabric Distributed Reduction

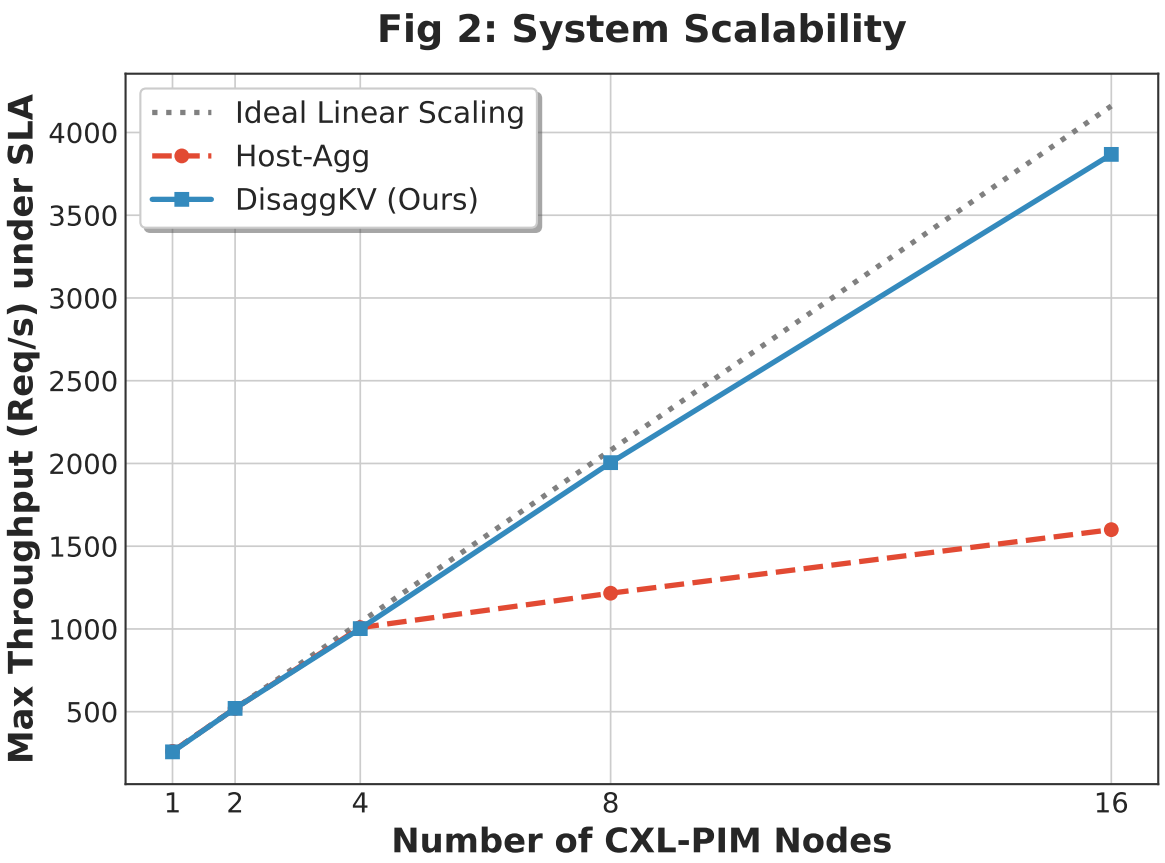
- **P2P Synchronization**: Two-Phase Commit communicating via switch-level P2P links.
- **Process**:
  1. Local assessment of maximum logit (`PIM_SEND_PARTIAL`).
  2. Global Max Distribution asynchronously.
  3. Global Summation accumulation.
- **Effect**: Exchanging kilobytes of scalars instead of gigabytes of matrices.
- **Resiliency**: Async FIFOs absorb fabric jitter and random delays.

Evaluation: Throughput SLA Compliance



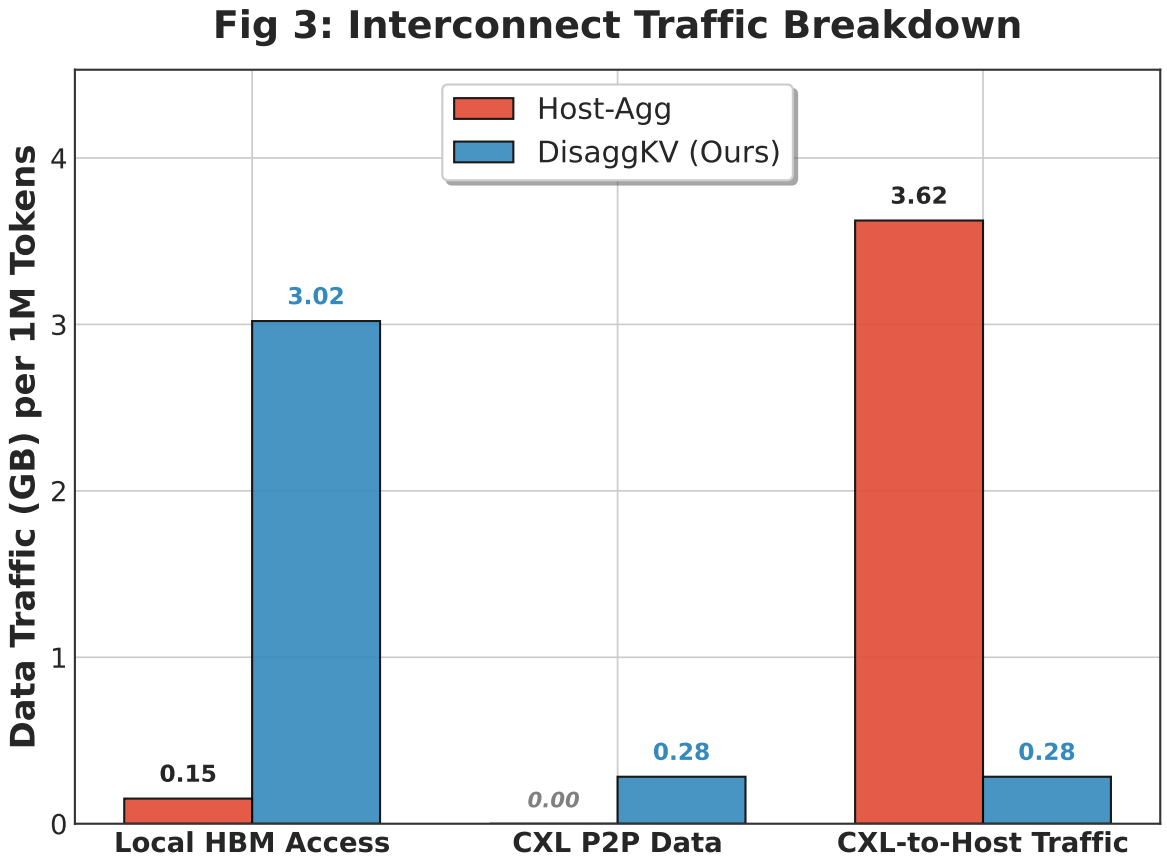
- DisaggKV maintains stable tail latency (<**15ms**) up to 3000 req/s.
- Host-Aggregated violates 50ms SLA at 600 req/s.
- **5x Throughput Improvement**.

Evaluation: Topology Scalability



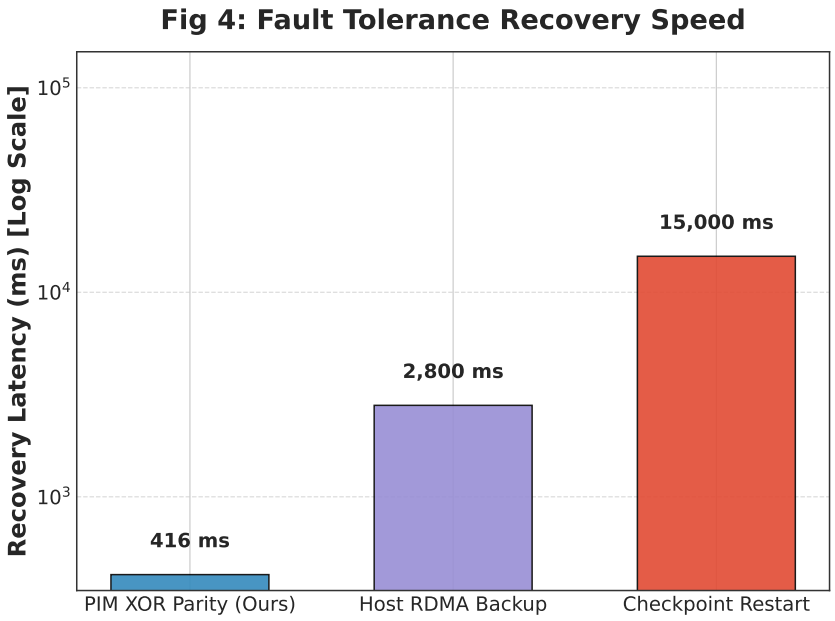
- **Near-Linear Scaling**: Near-ideal horizontal expansion up to 8 nodes.
- Host-Aggregated yields diminishing returns due to Root Complex port contention.

Evaluation: Bandwidth Reduction

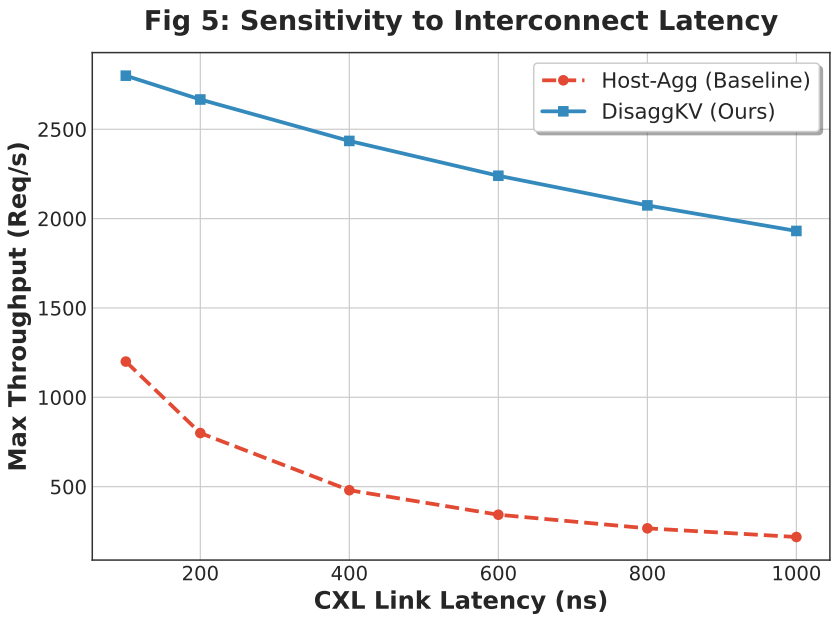


- Global switch traffic reduced by over **92%**.
- Replaces dense matrix transfers with condensed scalar operations.

Evaluation: Fault Recovery Sensitivity



Sub-second restoration (P99 < 500ms).



Resilient to network diameter/latency increase.

- PIM-side XOR parity recovery vs. multi-second Host software restarts.
- Resilient to large-scale data center cable lengths/hops.

Conclusion

- **Capacity vs. Bandwidth**: DisaggKV solves the bottleneck in clustered CXL environments.
- **Hardware-Software Co-Design**: Hypervisor paging policy + CXL 3.0 P2P fabric.
- **Key Outcomes**:
  - 92% reduction in global switch traffic.
  - 5x throughput gain under strict SLA.
  - Sub-second fault recovery.
- **Conclusion**: Critical enabler for large-scale multi-tenant LLM serving.

Thank You!

Questions?